Manning manual of style

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NOTE This guide doesn't have an index. To quickly locate any topic, use the Find feature in Acrobat: press Ctrl-F (or Command-F) and enter your search term.

The following guidelines should be used for all Manning books. **These rules are somewhat mutable depending on the topic, the type of book, and author preference.** Check with Alisa (alla@manning.com) or your production editor for any clarification needed.

1 Important general points for Manning books

- During editing, the emphasis should be on making sure text is readable, clear, and correct. **In addition to ensuring that the basics of grammar and punctuation are correct, be proactive about editing text that is clumsy or difficult to read. Break up long and complicated sentences as needed. If a sentence's structure or meaning is unclear or hard to follow, query the author to reword or reword and query for their approval.** Ideally, the author's writing style should remain intact as much as possible—for example, follow the author's preference when it comes to voice ("we" vs. "you"), use of contractions, and casual vs. more academic presentation style—but the priorities are clarity and correctness.
- Manning uses **inclusive language** that is free from words, phrases, or tones that reflect prejudiced, stereotyped, discriminatory, or limited views of particular people or groups. Avoid references to personal attributes such as age, gender, sex, race, ethnicity, culture, sexual orientation, and disability unless directly relevant to the topic being discussed.
- Our books are technical, and our audience is global and diverse. So, as a matter of policy, we **avoid political or religious comments** in our books, including acknowledgments and dedications.
- Our books use **American English** rules for spelling, punctuation, and style.
- References to people should be **gender neutral**. Use *them*, *they*, and *their* in the singular form; do not use *he/she* or *him/her* or alternate between masculine and feminine pronouns.
- **Check the spelling and capitalization of all names** of companies, products, and individuals. Many companies and products use CamelCase: words and phrases that are written without spaces but often with internal capitalization. Do not rely on the author spelling any name correctly—authors often misspell names or spell the same name multiple ways (for example, Java Script, Javascript, JavaScript). Use Google to check the names, verify them directly at a company or product website if possible, and add them to the book's stylesheet.
- **Companies and other organizations use singular pronouns**: "Microsoft released its newest product"; "The company is moving its headquarters to Wichita."
- Active passive voice and present tense are preferred. The voice and tense should also be consistent. If a book is primarily in passive voice or a different tense check with Alisa (alla@manning.com) before recasting.
- **Don't use one-sentence paragraphs**—merge them into the surrounding text.
- **Break up long paragraphs** to avoid a"wall of text"
- Use one space at the end of a sentence.
- All paragraphs should¶ end with a *hard return* (created by pressing the Enter↵ key). The paragraph mark looks like a backward P: ¶. If a paragraph mark looks like a left-pointing arrow ↵, it is a soft return (created by pressing Shift-Enter) and should be replaced by a hard return.
- Manning books **do not use Wikipedia as an information source** (except in very rare cases, requiring approval; check with Alisa) and ***never* use photos, diagrams, or other figures cited from Wikipedia**. For specific information on how Manning deals with Wikipedia, see section 23.
- If the text refers to **colors** shown in figures, query the author to reword to remove them. Almost all Manning books are printed in black and white, and the parts of a figure must be identifiable without references to colors.
- For grammatical issues not covered by this style guide, refer to the Chicago Manual of Style. For spelling, refer to Merriam-Websters.

Other points to remember

- ⇨ If you have questions or concerns during a project, contact your production editor and/or Alisa Larson (alla@manning.com).
- ⇨ If a book strays from the style outlined herein, check with your production editor and/or Alisa on how to proceed.
- ⇨ Never make significant changes to a book's style or structure without getting confirmation first.
- ⇨ **Copy your production editor on all communications with authors.**
- ⇨ Keep the Production sheet up to date.
- ⇨ Submit final edited or proofread files individually or in batches.

1.1 Forbidden words

Per Manning's publisher, search for and replace the following words in all books:

- *Impact*—Replace with *affect*, *influence*, or similar (unless the text is about hitting something and the resulting physical impact).
- *Issue*—When referring to something negative, replace with *problem* or similar.
- *Leverage*—Replace with *use* or similar (unless the text is saying that something is multiplied, like the force when using a lever).
- *Most simplistic*—Replace with *simplest*.
- *Simplistic*—Replace with *simple* or *overly simple*.
- *Remediate*—Replace with *fix*, *correct*, or similar.
- *Variance*—Replace with *variety*, *variability*, or similar (unless the text is about statistics and the actual measure called the variance).
- *Verbosity*—Replace with *verboseness*.

1.2 Suggested beginning workflow for copyediting

The following provides a suggestion workflow as you begin editing a chapter. Your workflow may vary but should cover all the tasks listed here:

1. Open the chapter file.
2. Turn on Show Paragraph Marks and Other Hidden Formatting Symbols (Ctrl-Shift-8) so you can easily see spaces, paragraph marks, tabs, etc.
3. Check to be sure the chapter uses only Manning Word styles to format text; fix any styles/formatting as needed (see section 2.2).
4. Do a search for double spaces in text and replace them with single spaces. (Don't use Replace All, because that will change spacing in code.)
5. Search for tab characters (enter ^t in the Find What box). If you find any, query the author to replace them with spaces.
6. Search for and replace the words listed in section 1.1
7. Search to be sure that all figures and tables are mentioned in the text (see section 16.1) and that all figures, tables, and code listings are in numerical order.
8. Check to be sure that all heading levels are nested correctly: see section 8.
9. Download the free version of Grammarly (https://grammarly.com) and use the tool to do a quick initial check of spelling, punctuation, and other issues.
10. Run Word spellcheck (in addition to Grammarly). Also be sure to run spellcheck as your very last step before submitting final chapters for typesetting.

If working in another format (ASCII or LaTeX), email Alisa Larson (alla@manning.com) for guidance if needed.

> **NOTE** All copyeditors are expected to download, install, and use the free version of Grammarly (https://grammarly.com). If editing in a format other than Word (LaTeX or ASCII), copy and paste text into Word to run Grammarly and spellcheck. (You don't need to make all changes suggested by Grammarly.)

2 Formatting text

The text should be formatted when you receive the files to be edited. However, almost all books will need some adjustment (add/remove/change) to the formatting. If you receive a project that is not formatted or has significant formatting problems, contact your production editor.

All text formatting in Manning books is applied using styles, *not* by selecting formatting buttons on the Word Ribbon. It's important for ease of typesetting that books use *only* Manning styles. The Manning template uses two kinds of styles:

- *Paragraph styles* apply formatting to all the text in a paragraph. For example, choosing the .Head 1 style formats the entire current paragraph as 12-point italic, bold Arial font with a reddish color; it also automatically assigns the next section number to the paragraph. As another example, choosing the .Code style formats the entire current paragraph as 8-point Courier New, with automatic spell-check turned off.
- *Character styles* apply formatting only to selected text. For example, selecting a block of text and choosing the .Italics style italicizes that text, and choosing .Code in Text formats the text as 9-point Courier New. If the style show in the Styles menu with a + sign (e.g., Body 1 + Italic), it needs to be restyled without using Word's functions (.Italics).

> **NOTE** Character styles cannot be "stacked" on each other: only one character style can be applied to any given text. For example, if you format some text using .Italics and then format the same text using .Bold, the result will be bold text, not bold italic text. To format text as bold italic, you need to use the .Bold Italics style.

Table 1 lists the Manning styles.

> **NOTE** If an author requests formatting that is not represented by one of the standard Manning styles and that is not noted in the handoff document, ask your production editor how to proceed.

Table 1 Paragraph and character styles used in the Manning template

Paragraph styles			Character styles	
.Body	.Head 1	.Sidebar	.Bold	.Code Underline
.Body 1	.Head 2	.Sidebar End	.Bold Italics	.Equation Caption
.Callout	.Head 3	.Sidebar Head	.Callout Head	.Hyperlink
.Code Annotation	.List Body	.Table Body	.Code	.Superscript
.Code Listing Caption	.List Bullet	.Table Caption	.Code in Table	.Subscript
.Equation	.List Bullet Sub	.Table Footer	Code in Text	.Underline
.Figure	.List Numbered	.Table Header	.Code Bold	
.Figure Caption	.List Numbered Sub	.Typeset Note	.Code Italics	
.Glossary Definition	.Quote	CO Chapter Number		
.Glossary Term	.Quote Source	CO Chapter Title		

Some elements of a chapter use *only* the formatting built into the paragraph style; other text can have a character style applied to add formatting in addition to that of the paragraph style.

Table 2 summarizes where character styles can be used.

Table 2 Paragraph style only vs. paragraph plus character styles

Chapter elements	
Only* paragraph styles	**Paragraph plus character styles**
Chapter title	Body text
Section headings	Captions (figures, tables, and code listings)
Callout and sidebar headings	Callout and sidebar text
Table column headings	Table body text
Code annotations	Code snippets and listings
	List text

*Chapter elements in the left column should never include .Italics, .Code, .Code in Text, .Bold, etc.

2.1 Text formatting conventions

Following are guidelines for where various types of text formatting are used in Manning books. Some formatting is included in the paragraph style applied to the text; other formatting is added to selected text using a character style.

> **NOTE** Table 2 shows where character styles may and may not be used: in case of a conflict between the following list and that table, the table rule takes precedence. For example, a code function in a text paragraph is formatted using .Code in Text, but the same function in a section heading or code annotation does *not* use code font—it is the same format as the rest of the text.

- Regular text font—Filenames, file paths, file extensions, database elements (table names, column headings, and so on), fields, and labels. Some authors prefer to use code font for some or all of these items; if so, be sure the preferred style is consistent throughout, and make a note in the book's stylesheet.

> **NOTE** The default style for filenames, file paths, folders, file extensions, etc. is regular font. However, if an author requests one of these text elements in code font, it's OK; just be consistent.

- *Italics*—New terms, emphasized terms, variables used in mathematical equations. Occasionally used to distinguish lines of code output from lines of input (.Code Italic).
- **Bold**—Rarely used in text. Avoid if possible and use only if there's a good reason italics can't be used instead. Bold is more commonly used to distinguish new or revised code in a snippet or code listing (.Code Bold).
- `Code`—Text the reader enters and elements of programming languages or of programs. Use .Code *for code listings and snippets* (code that appears on its own line), .Code in Table *for code terms in tables,* and .Code in Text *for code terms anywhere else* (e.g., code on the same line with other text, in figure and listing captions, etc.) The elements that appear in code font may vary between books, depending on the author's preferences; they may include classes, namespaces, interfaces, packages, methods, functions, statements, formulas, prompts, macros, data types, properties, buffers, and so on.
- Blue underlined—URLs and email addresses (.Hyperlink).

2.2 Checking and fixing formatting

For various reasons, chapters may include extraneous or incorrect formatting. For example, the author may apply formatting such as italics or code font manually rather than using the appropriate Manning style—a difference that isn't visible but that matters when the text is imported into the typesetting program.

> **NOTE** If you have Show Hidden Text active (Ctrl-Shift-8), the first page of a chapter may have a list of unnecessary additional chapter numbers. The typesetter will correct the numbering. You don't need to worry about it.

This section explains the quickest way to check the formatting of a chapter and fix styles as needed. Suppose you have inserted the example text shown in figure 1 into the Manning template. All the formatting *looks* correct—but you need to check the styles to be sure.

Figure 1 All the formatting in this text looks correct. But was it applied manually or with Manning styles?

2.1 → Alpakka·Kafka¶

Alpakka·Kafka·is·built·on·Kafka·Streams.·In·this·chapter,·you·use·the·Jackson·library—a·JSON·parser—for·serialization,·so·the·data·in·these·examples·is·stored·as·JSON·in·Kafka.·¶

The·`groupId`·is·used·to·track·consumption·of·the·topic.·Consumption·is·stored·in·what·is·called·the·*offset*,·and·each·`groupId`·has·its·own·offset.·As·soon·as·a·batch·of·elements·from·the·topic·is·consumed,·a·new·offset·is·stored.·This·marks·the·last·read·position·in·the·topic·for·that·`groupId`.·This·action·is·called·*committing·the·offset*.·By·default,·the·Alpakka·plain·consumer—`plainSource`—does·not·commit·this·offset.·When·the·application·restarts,·it·does·not·pick·up·where·it·left·off.¶

NOTE·For·more·information,·see·https://kafka.apache.org/documentation.¶

The·`config`·points·to·the·default·Kafka·Streams·configuration,·and·you·need·the·`ActorSystem`·to·retrieve·it:¶

```
object·ConsumerPlainSource·{¶
··implicit·val·system·=·ActorSystem(Behaviors.empty,·"consumerOne")····#A¶
··val·config·=·system.settings.config.getConfig("akka.kafka.consumer")¶
}¶
```

#A·Implicit·system·to·provide·the·materializer·to·the·stream¶

STEP 1: DISPLAY ALL THE FORMATTING USED IN THE DOCUMENT

To see what styles and formatting are currently in use in a chapter, do the following:

1. Click the Home tab on the Ribbon. In the Styles group, click the arrow in the lower-right corner.

   When you do, the Styles pane opens, as shown in figure 2, listing all the available styles. You need to change this display so that it shows the styles and manually applied formatting currently being used in this document.

2. At the bottom of the Styles pane, click Options. In the Style Pane Options dialog box, choose these settings:
   - Select Styles to Show > In Use.
   - Select How List Is Sorted > Alphabetical.
   - Select Formatting to Show as Styles check boxes (figure 3).

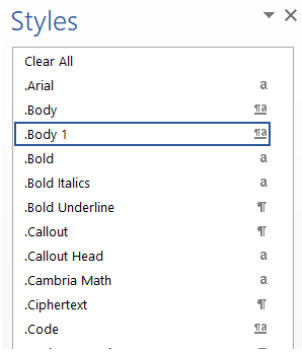


Figure 2 The Styles pane displaying all available styles in a document

3. Click OK. The Styles pane now lists only the styles and added formatting that are in use in the document at this moment.

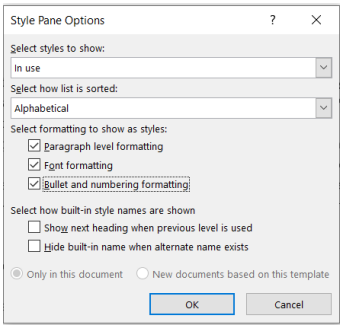


Figure 3 Options to display all styles and manual formatting currently in use

> Select all three

Figure 4 shows the Styles pane for the example text in figure 1. Anything after a plus sign (+) is formatting that has been applied manually rather than by using a style.

STEP 2: SELECT TEXT BY FORMAT, AND APPLY MANNING STYLES

You're now ready to select instances of incorrectly applied formatting in the Styles pane and apply the correct styles. The Styles pane always includes the Clear All option and the Default Paragraph Font item, so you can ignore them. **Look for any non-Manning styles—that is, any styles not listed in table 1—and any manually applied formatting indicated with a +.** Figure 4 shows that the example text includes four kinds of manual formatting and a non-Manning style that need to be fixed.

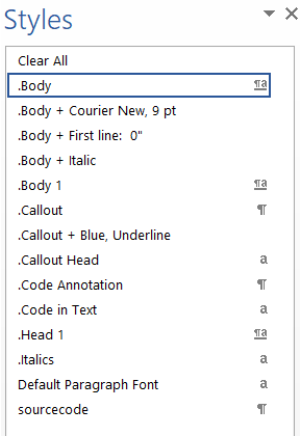


Figure 4 Styles and manual formatting (anything after a +) used in the text shown in figure 1. Although all the formatting in the example looks correct, the Styles pane shows that the author used manual formatting and a non-Manning style.

It's easiest to start at the top of the list in the Styles pane and work your way down:

1. Right-click the first item that needs to be corrected. In the example, it's *.Body + Courier New, 9 pt*. A pop-up menu like the one in figure 5 opens.

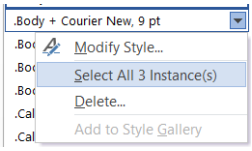
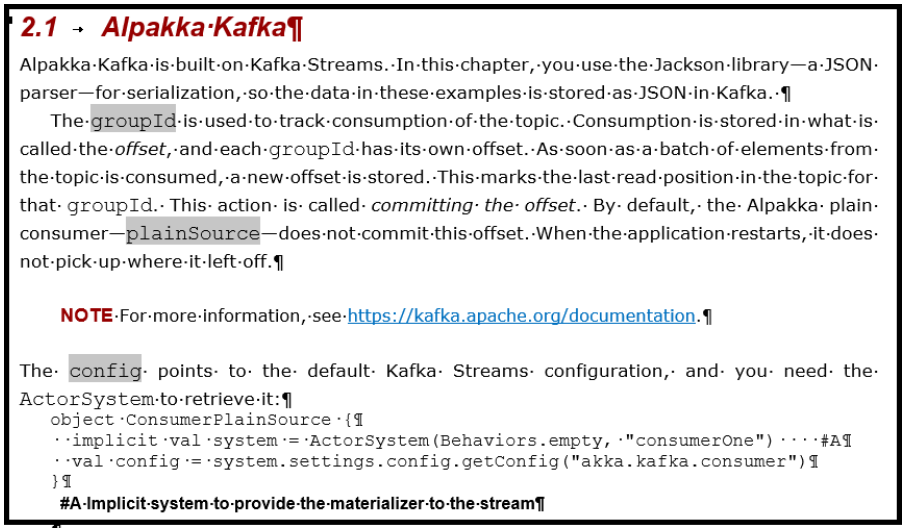


Figure 5 Right-click an item that needs to be fixed.

2. Choose the menu option that says Select All *X* Instances. Word will select all instances in the document of text that uses the selected style/formatting. If instances appear on multiple pages, Word will select all of them and display the last instance in the document.

   In the example document, Word selects the three instances where the author manually applied 9-point Courier New font to text instead of using the Manning .Code in Text style (figure 6).



2.1 → Alpakka·Kafka¶

Alpakka·Kafka·is·built·on·Kafka·Streams.·In·this·chapter,·you·use·the·Jackson·library—a·JSON·parser—for·serialization,·so·the·data·in·these·examples·is·stored·as·JSON·in·Kafka.·¶

The·`groupId`·is·used·to·track·consumption·of·the·topic.·Consumption·is·stored·in·what·is·called·the·*offset*,·and·each·`groupId`·has·its·own·offset.·As·soon·as·a·batch·of·elements·from·the·topic·is·consumed,·a·new·offset·is·stored.·This·marks·the·last·read·position·in·the·topic·for·that·`groupId`.·This·action·is·called·*committing·the·offset*.·By·default,·the·Alpakka·plain·consumer—`plainSource`—does·not·commit·this·offset.·When·the·application·restarts,·it·does·not·pick·up·where·it·left·off.¶

NOTE·For·more·information,·see·https://kafka.apache.org/documentation.¶

The·`config`·points·to·the·default·Kafka·Streams·configuration,·and·you·need·the·`ActorSystem`·to·retrieve·it:¶

```
object·ConsumerPlainSource·{¶
··implicit·val·system·=·ActorSystem(Behaviors.empty,·"consumerOne")····#A¶
··val·config·=·system.settings.config.getConfig("akka.kafka.consumer")¶
}¶
```

#A·Implicit·system·to·provide·the·materializer·to·the·stream¶

Figure 6 Word selects the instances that have the chosen formatting.

3. Determine which Manning style should be applied to the selected text. In this case, the text should be assigned the .Code in Text style. With the text still selected, click the correct style in the Style Gallery on the Styles tab of the Ribbon (figure 7).

   Word assigns the Manning style to all selected text. The non-Manning style/formatting item disappears from the list in the Styles pane.

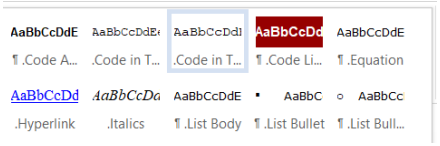


Figure 7 Click the correct style in the Style Gallery.

4. In the Styles pane, right-click the next style/formatting item that needs to be corrected. In the example, it's *.Body + First Line: 0"*. Choose the Select All *X* Instances option (figure 8).

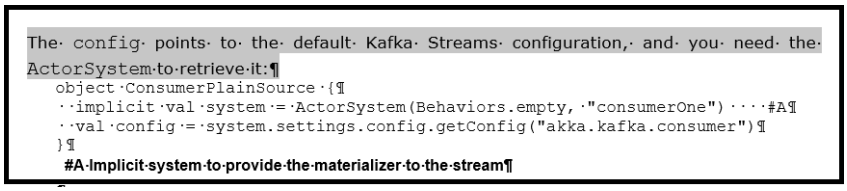


Figure 8 Word selects the next instance to be fixed.

As before, determine which Manning style should be applied to the selected text. In this case, the author has manually removed the indent from the .Body style. The text should be assigned the .Body 1 style, instead.

5. Choose the appropriate Manning style from the Style Gallery to apply it to the selected text.

Repeat this process until you've replaced all non-Manning styles and formatting with Manning styles. In the example, these additional changes need to be made:

- Selecting all instances formatted with *.Body + Italic* and applying the .Italics style
- Selecting all instances formatted with *.Callout + Blue*, Underline and applying the .Hyperlink style
- Selecting all instances formatted with the non-Manning *sourcecode* style and applying the .Code style

What is Default Paragraph Font?

The Styles pane lists a style called Default Paragraph Font. If you select all instances of the style, it applies to the entire document. That's because it refers to the font that goes with the style applied to any given paragraph.

Although Default Paragraph Font applies to every paragraph, the default font varies from paragraph to paragraph. If, for instance, a word in a .Body para has incorrect character/font formatting, you can select it and apply Default Paragraph Font, and the word will go back to the .Body font: the default font of the paragraph's style.

3 Capitalization

- Use **sentence capitalization** (only the first word and proper nouns uppercase) for chapter titles, section headings, and captions. Also use sentence capitalization for labels, annotations, and explanatory text in figures.
- Use **lowercase** for frontmatter and backmatter titles ("preface"; "appendix A").
- Use **lowercase** for references to chapters, figures, listings, and tables ("as shown in figure 3.6").
- Use **title case** (first word, last word, and major words uppercase) for onscreen elements: dialog box titles, items chosen from list boxes, check box options, and so on (e.g., Choose the Bullet option from the Choose Format for This Level menu).
- If a book contains references, follow the author on capitalization of titles, but make consistent if necessary.
- For mentions of books and articles in the text, use title case:
    - To learn more, see JJ Geewax's *API Design Patterns* (Manning, 2021).
    - If you are interested, check out Robin Gandy's "What Is Web Design" (https://www.pagecloud.com/blog/web-design-guide).

4 Punctuation

4.1 Commas and periods

- Use series commas (red, white, and blue).
- When *independent clauses* are joined by *and*, *but*, *or*, *so*, *yet*, or any other conjunction, use a comma before the conjunction: "You'll create a new app, and we'll explain how to format it."
- Use commas after opening dependent clauses, prepositional phrases, etc.
- If a figure or table caption is a complete sentence, use a period.
- If one or more items in a bulleted or numbered list are complete sentences, use periods for all items within that list, even fragments.
- If a code annotation is a complete sentence or a mix of fragments and sentences, use a period for sentences only. Annotations are not treated like lists; it is fine for some annotations to be complete sentences and end with periods while others are phrases with no period.
- If a table cell or figure contains a complete sentence or a mix of fragments and sentences, use a period for sentences only. Table rows and columns are not treated like lists; it is fine for some cells to be complete sentences and end with periods while others are phrases with no period.

4.2 Apostrophes and quotation marks

- Punctuation goes *inside* quotation marks unless the text in quotes is an actual value, and it must be clear that sentence punctuation isn't included.
- **All quotes/apostrophes should be curly in text and straight in code snippets/listings and code elements in text. Replace as necessary.**

4.3 Colons

- *In text*, use a lowercase letter after a colon unless the colon introduces a series of related sentences or questions. *In a chapter title or heading*, capitalize the word after a colon (because what comes after the heading is considered a subtitle).
- Use a colon, not an em dash, to separate parts of a chapter title or heading. For example: "Introduction to widgets: Building them from scratch."
- Whether to use a colon before a code snippet or a numbered or bulleted list depends on whether the lead-in to the code is a complete sentence. If it is, you need a colon:

  The following code snippet shows an example of what we mean:

  ```
  my_object + my_other_object
  ```

  You can write code like this in many situations.

  If you write code like

  ```
  my_object + my_other_object
  ```

  then you have a multipurpose tool good for many situations.

  In the first case, a complete sentence leads up to the code. So, the sentence ends with a colon. The text following the code begins a new sentence. In the second case, the code is part of the sentence. So, the lead-in phrase doesn't get any punctuation, and the text after the code begins with a lowercase letter because it continues the sentence.

4.4 Hyphens and dashes

- Hyphens are used in compound adjectives ("a short-term result") and in keyboard shortcuts (Ctrl-Alt-Delete).
- En dashes (–) are used to indicate a range. Use the word "to" in the main sentence ("The code will take 2 to 14 seconds to run."). Use an en dash in parentheses [The code will take some time to run (2–14 sec)"], table bodies, and page ranges. Also use an en dash for modifiers of compound words, whether or not they are hyphenated:

- personal computer–based software
- privacy-preserving–based algorithms
- Amazon S3–related programs

- Em dashes (—) indicate a break in a sentence: "While the code is running—and it may take a while—go get some coffee."

4.5 Ellipses

When an ellipsis is used to indicate the omission of text or code, use three spaced periods (nonbreaking spaces [Ctrl-Shift-Spacebar] before, after, and between the periods: °.°.°.°). Add a period *before* an ellipsis to indicate the omission of the end of a sentence or the omission of text after the end of a sentence. Other punctuation can precede or follow an ellipsis as needed. Text before and after an ellipsis should form grammatically complete sentences. The first word after an ellipsis is capitalized if it begins a new sentence.

Delete ellipses if they are used for a continuation of thought ("We'll discuss that in the next chapter. . . .") or emphasis ("The answer is . . . 5.257.").

5 Dates and numbers

- Spell out numbers under 10. *But* if a sentence or paragraph includes multiple numbers in the same category, and any of those numbers is 10 or greater, write all numbers in that category as numerals. For example, "A team of seven people is generally enough to corral 12 cats and 3 dogs."
- For units of *measurement*, use numerals: 5 inches, 2 cm, 6 feet, 2 bytes.
- For units of *time in hours or less* (hours, minutes, seconds, milliseconds, etc.), use numerals: 2 hours, 9 min, 1 ms, 7 seconds.
- For units of time in days or longer (days, week, months, and years), spell out the number if under 10 (two days, one month, 26 years).
- When writing abbreviated units, leave a space between the number and the unit. Do *not* use a period after the abbreviation, except in the case of *in.* for inch (to avoid confusion). Examples: 64 MB, 256 Mbps, 50 MHz, 18 in., 50 m, 100 pt, 50 ft, 9 s (or sec; seconds), 4 min (minutes), 35 ms (milliseconds).
- Follow author preference for abbreviating units of measure and time. Either abbreviated or spelled-out units are fine; just make sure the usage is consistent. Never abbreviate days, weeks, months, or years.
- Use numerals for versions (Version 5 or v5).
- Use numerals for actual values ("If the `Height` property is set to `1`, click OK.").
- Use numerals with the percent sign ("A survey found that 87% of shops use Java.").
- 32-bit.
- 640 × 320 (put a space on either side of the ×).
- 1980s; the 20th century.
- Use a comma with thousands (1,000).
- 2 million, not 2,000,000.
- Use % (not percent). Use "to" in the main text, "85% to 99%," and an en dash in parentheses, tables, and figures (85%–99%). Exception: if beginning a sentence, spell out *percent*: "Twenty percent of the total …"

6 Menu selections, keypresses, and onscreen options

When the reader is instructed to make a choice from a menu or set of menus, use the following format:

- Choose Tools > Macro > Record New Macro.
- Dialog box titles, onscreen options, items chosen from list boxes, check-box options, and so on should be written using title case, to avoid confusing the words with the surrounding text. An example is
- You will see the option to Add to Cart. Click it and then scroll down to find Continue to Checkout.

- Use this capitalization even if the options appear in sentence case on the actual screen. Do not use quotation marks.
- When the reader is asked to use keys on the keyboard, the verb is *press*. When the reader needs to interact with an onscreen button, the verb is *click* (not *click on*).

7 Chapter-opener bullets

In most Manning books, chapters begin with a list of bullets in a shaded box, introducing the content covered. (There are exceptions, depending on the book series.) This list appears about halfway down the first page in a shaded box; see figure 9 for an example. **To ensure that there is room below the box to start the chapter text, the list should ideally consist of three to five bullets totaling no more than eight lines** (each line can fit a maximum of about 45 characters).



Figure 9 An example of the list of chapter-opener bullets that appears on the first page of each chapter in many Manning books. Each typeset line can have at most about 45 characters. Ideally, the list includes three to five bullets and no more than eight lines, to leave plenty of room on the page for the chapter text to begin.

Following are the guidelines for these lists:

- The list is introduced by the phrase "This chapter covers" with *no colon* after it.
- List items should be short and snappy.
- Recommended number of bullets: 3 to 5.
- Maximum characters/spaces per typeset line: 45.
- Maximum lines per bullet: 2.
- Maximum total number of lines in the list: 8.
- Maximum number of characters/spaces in the list: 360.

9 Summary sections

In almost all Manning books, each chapter ends with a section titled "Summary." This section has a level-1 heading but no section number, and it doesn't appear in the book's TOC. It consists of a list of bullets reiterating the important concepts covered in the chapter. Keep in mind the following:

- There's usually not a lead-in paragraph, but if it's there, leave as is. However, **no text/paragraph should appear *after* the list**. Query the author to remove. If an author insists it stays, let Alisa (alla@manning.com) and your production editor know.
- **The Summary section must be the very last section of each chapter**. For example, if a chapter includes an Exercises section, that section must appear before the Summary section.

EXCEPTION If the author uses numbered citations and the list of numbered references appears at the end of each chapter, place after the Summary section (see section 17.1).

- If a Summary section is missing or deviates from this format, notify Alisa and your production editor.

Each item in the Summary list should be a complete sentence or two, reminding the reader of what they've learned and perhaps offering additional insight. If an author provides new information not covered in the chapter, discusses upcoming chapters, etc., query them to provide summary items instead.

10 Appendixes

Appendixes appear at the end of a book and provide supplemental information. The word *appendix* in the title is lowercase. **An appendix does not have This chapter covers or a Summary section.** The following notes apply to appendixes:

- Appendixes are lettered (A, B, C), not numbered. If a book has only a single appendix, it doesn't need a letter; it's just the appendix.
- Headings in appendix sections may or may not be numbered, depending on the content and author preference.
- Figures and tables in the appendix are numbered starting with a capital *A* (e.g., figure A.1, A.2, etc.) for appendix A, capital *B* for appendix B (e.g., table B.1, B.2, etc.), and so on.
- Use lowercase when referring to the appendix in the text (eg, "The directions for using JavaScript are in the appendix" or "The directions for using JavaScript are in appendix A.").
- **Be sure that all appendixes are called out somewhere in the text at least once.** For example, if appendix B is provided, make sure the author refers to it somewhere in the book.
- **Be sure that an appendix is provided for each callout.** For example, if the author calls out appendix C, make sure appendix C exists.

8 Chapter titles and section headings

The Manning template uses automatic numbering for the **CO Chapter Number**, **.Head 1**, and **.Head 2** styles. The **.Head 3** subhead style isn't numbered. For more on autonumbering, see ***Using the Manning template.***

Chapter headings are numbered within a chapter as follows:

13	.CO Chapter Number	Capitalize like this: The chapter title
13.1	.Head 1	Capitalize like this: An L1 section title
13.1.2	.Head 2	Capitalize like this: An L2 section title
[no number]	.Head 3	Capitalize like this: An L3 section title

The first section heading should come within the first two pages of the chapter. The introduction (which never uses a header) should not contain substantive material such as figures or tables. If no heading appears in the first two pages, query the author to provide.

Follow these guidelines:

- Manning books don't use more than three heading levels.
- Headings can't be "stacked": there should always be text between two headings.
- A heading must always be followed by at least a sentence of regular text—it can't be immediately followed by a list, code, callout, etc.
- There should be more than one heading at any given level within another level. That is, a chapter must have more than one level-1 heading. Within each level-1 section, there can be zero level-2 headings or two or more level-2 headings, but there can't be just one level-2 heading. And within each level-2 section, there can be zero or two or more level-3 headings; there can't be just one.
- All title and heading text uses the same font and style; titles and headings don't include code font, italics, or other formatting. That is, they do not use character styles (see Table 1).
- Use sentence capitalization for chapter titles and section headings.
- Section headings never end with a period, even if a heading is a complete sentence.
- In a chapter title or heading, capitalize the word after a colon (because what comes after the heading is considered a subtitle).
- Use a colon, not an em dash, to separate parts of a chapter title or heading. For example: "Introduction to widgets: Building them from scratch."
- Section headers should not define abbreviations (see section 26.2).
- Section headers should not consist solely of an abbreviation (see section 26.2).

11 Source code

Code is set in `fixed-width font`, both on standalone code lines (formatted with .Code) and when code terms are used in text (formatted with .Code in Text). Use .Code in Table for text in tables. Standalone code may appear in a listing with a number and a caption or may be an unnumbered snippet:

- *Code snippet*—A section of code that appears between paragraphs of text and has no number or caption. Code snippets use the .Code style.

Snippets are (usually) used for short bits of code or commands that are being discussed in the surrounding text. Although they appear to be standalone page elements, they should read as though they're parts of the paragraphs that surround them. They can be used for command-line commands, command and program output, syntax illustrations, and one-time general examples.

If the text leading into a code snippet is a complete sentence, it should end with a colon. If the lead-in is a phrase, not a complete sentence, it doesn't get any punctuation (see section 4.3).

- *Code listing*—A section of code (formatted with the .Code style) with a bar at the top containing a number and caption. The number and caption together must fit on a single line (65 characters or fewer), the caption should not be a complete sentence, and it does not end with a period.

  Listings are (usually) included in a book's downloadable source code. They're related to the chapter section in which they appear, but they're more standalone than snippets. They may serve as goals, recaps, summaries, or conclusions to a section. Listings tend to be longer than snippets, but that isn't always the case.

The following guidelines are important to accommodate our typesetting software and the size limits of the printed page:

- **Use spaces, not tabs or margin settings, to indent code.** Usually, each tab can be replaced with four spaces (but some authors may prefer a different number of spaces). If you are unsure, query the author.
- **A line of code can have no more than 76 characters. If the line has a code annotation next to it, it should have no more than 55 characters.** If a line is longer than these limits, the line should be rewritten or manually broken. If necessary, a "[CA]" tag for a code-continuation character can be added at the start of a wrapped code line (the typesetter will replace it with an arrow (➥)):

```
    log.debug "Incoming Queue Request from:
[CA]${msg.userId} to: ${msg.to} content: ${msg.content}"
```

- **Do not change code. Always query the author to fix typos and/or break long code lines.**

EXCEPTIONS Replace tabs with four spaces. Change curly single and double quotes in code to straight quotes.

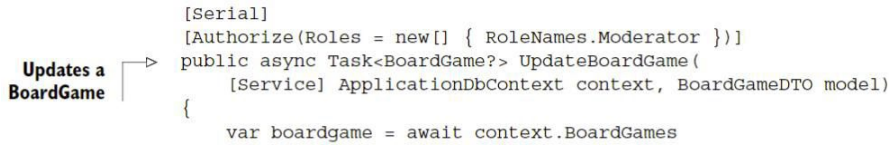
12 Code annotations

The term code *annotation* refers to a comment placed on either side of a code line on the printed page to explain what's going on at that spot in the code. Annotations are different from comment lines that are part of a code listing—**Manning books use annotations instead of comment lines**. (The exceptions are comments that are an integral part of code being written as part of an example.)

An annotation (styled .Code Annotation) in the Word file will look like this:

```
········[Serial]¶
········[Authorize(Roles·=·new[]·{·RoleNames.Moderator·})]¶
········public·async·Task<BoardGame?>·UpdateBoardGame(·#A¶
············[Service]·ApplicationDbContext·context,·BoardGameDTO·model)¶
········{¶
············var·boardgame·=·await·context.BoardGames¶
#A Updates a BoardGame
```

In the typeset page, it will look like this:



Keep the following in mind when working with code annotations:

- Annotations can be phrases or complete sentences. End complete sentences with a period; sentence fragments don't need punctuation.
- All annotation text uses the same font and style; annotations don't include code font, italics, or other formatting.
- If an annotation is not a complete sentence, then in general, it should begin with an "-s" verb, so it describes what a line of code does ("Defines the Widget class"; "Prints the output"). If the author prefers active voice for annotations ("Define the Widget class.") that's fine, but be sure the style is consistent throughout.
- Don't shorten annotations by removing articles. Long annotations are OK. For example: "Calls the xyz() method" rather than "Calls xyz() method".
- If a code line has an annotation, the line should have no more than 55 characters, to allow space for the annotation on the printed page.
- If a code listing has annotations on several lines in a row—particularly if the code lines and/or the annotations are long—query the author to add blank lines in the code to make space for the annotations on the page.
- Be sure that the annotations letters in the code and the annotations given match. For example, if the author calls out #A, #B, and #C in the code lines, be sure all three annotations are listed below the code. Likewise, if the author provides three code annotations, be sure that #A, #B, and #C are called out in the code lines.

For more information about annotations, including examples and usage guidelines, please see the document ***Code annotation guidelines***.

12.1 Cueballs

Cueballs are number icons that appear in the margins of the code to identify a section of code that is discussed in the text. Cueballs have two main purposes:

- To allow for longer discussions of specific code line(s) in the text, and
- To allow specific code line(s) to be identified easily in the text.

Cueballs in code snippets and listings can be standalone (number only) or combined with an annotation, or a single listing can mix cueballs-only, annotations-only, and cueballs-with-annotations lines. A line of code only needs a cueball if it is explicitly mentioned in the text.

Cueballs alone

The mention or discussion of the code line (or lines) is placed solely in the text.

```
> y <- matrix(1:20, nrow=5, ncol=4)
> y
     [,1] [,2] [,3] [,4]
[1,]    1    6   11   16
[2,]    2    7   12   17
[3,]    3    8   13   18
[4,]    4    9   14   19
[5,]    5   10   15   20
> cells    <- c(1,26,24,68)
> rnames   <- c("R1", "R2")
> cnames   <- c("C1", "C2")
> mymatrix <- matrix(cells, nrow=2, ncol=2, byrow=TRUE,
                     dimnames=list(rnames, cnames))

> mymatrix
   C1 C2
R1  1 26
R2 24 68
> mymatrix <- matrix(cells, nrow=2, ncol=2, byrow=FALSE,
                     dimnames=list(rnames, cnames))
```

❶ Creates a 5 × 4 matrix

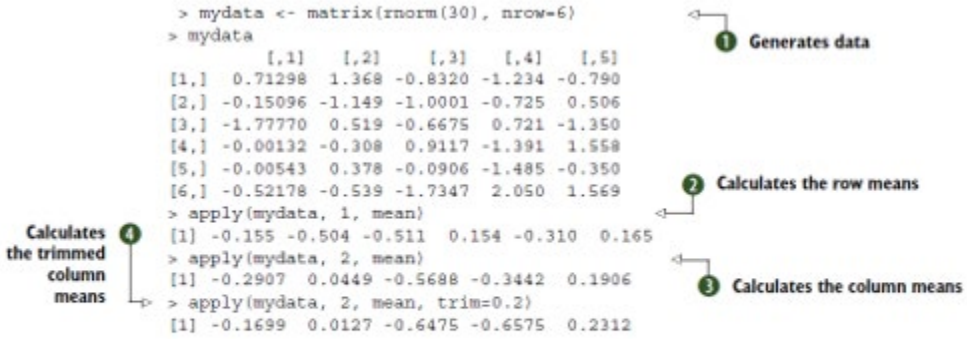
❷

❸ 2 × 2 matrix filled by columns

First you create a 5 × 4 matrix ❶. Then you create a 2 × 2 matrix with labels and fill the matrix by rows ❷. Finally, you create a 2 × 2 matrix and fill the matrix by columns ❸.

Cueballs with annotations

Brief annotations (max ≥75 characters) can be combined with cueballs. *For example,*

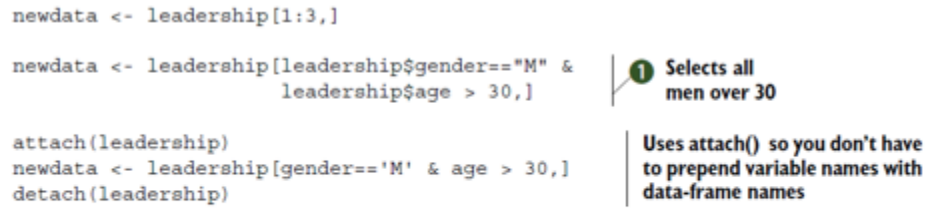


You start by generating a 6 × 5 matrix containing random normal variates ❶. Then you calculate the six row means ❷ and five column means ❸. Finally, you calculate the trimmed column means (in this case, means based on the middle 60% of the data, with the bottom 20% and top 20% of the values discarded) ❹.

TIP Be careful to avoid redundancy between the annotation and the text.

Cueballs and some annotations

Code can also use a combination of cueballs only, annotations only, and annotations with a cueball. *Remember, only code lines that are mentioned explicitly in the text need a cueball*:



Each of these examples provides the row indices and leaves the column indices blank (therefore choosing all columns). Let's break down the line of code at ❶ in order to understand it:

12.2 Using cueballs in text

Cueballs have just one firm rule: each cueball used in the code must have a corresponding mention in the text. Beyond that, Manning purposely does not have strict guidelines for using cueballs in the text. Authors should be allowed to use cueballs as they see fit, if their meaning is clear and consistent. Some of the most common ways to take advantage of cueballs are as follows.

Simple text mentions

Simple text mentions can be set anywhere in a sentence where they make sense. They are not a part of the grammatical makeup of the sentence. For example,

"You specify the desired mean vector and variance-covariance matrix ❶ and generate 500 pseudo-random observations ❷."

"For the second pie chart ❷, you convert the sample sizes to percentages and add the information to the slice labels."

Prepositional phrases

Cueballs are commonly used as nouns in prepositional phrases that describe placement. The cueballs are part of the grammatical structure of the sentence. For example,

"The code at ❶ produces cell frequencies for the three-way classification."

"The proportion of patients with None, Some, and Marked improvement for each Treatment × Sex combination is provided in ❹."

Sentence subject

In these cases, the cueball acts as the subject of the sentence and, thus, is part of the grammatical makeup of the sentence:

"❶ produces cell frequencies for the three-way classification."

Although this format is acceptable, Manning prefers a specific noun as the sentence's subject, such as

"The code at ❶ produces cell frequencies for the three-way classification."

Discussion beginners

Cueballs may be set at the beginning of a paragraph to indicate that the following discussion involves the code marked by the cueball. In this case, the cueballs are not part of the grammatical makeup of the sentence:

❶ The `quantile()` function gives you the percentile rank of each student's performance score. You see that the cutoff for an A is 0.74; for a B, 0.44; and so on:

```
> y <- quantile(roster$score, c(.8,.6,.4,.2))
> y
80% 60% 40% 20%
0.74 0.44 -0.36 -0.89
```

❷ Using logical operators, you can recode students' percentile ranks into a new categorical grade variable. This code creates the variable `grade` in the `roster` data frame:

```
> roster$grade[score >= y[1]] <- "A"
> roster$grade[score < y[1] & score >= y[2]] <- "B"
> roster$grade[score < y[2] & score >= y[3]] <- "C"
> roster$grade[score < y[3] & score >= y[4]] <- "D"
> roster$grade[score < y[4]] <- "F"
> roster
```

13 Figure, table, and listing captions

A *figure caption* follows the figure and is styled with .Figure Caption. A *table or code listing caption* appears at the top of the table or code listing (.Table Caption and .Code Listing Caption, respectively). A few notes about captions:

- Code listing captions must fit on one line, so they should be no longer than about 65 characters (including the listing number). They should *not* be full sentences and do not end with a period (e.g., change "Add method `xyz`." to "Adding method `xyx`").
- Figure and table captions can be any length and may or may not be full sentences. Use a period at the end of a caption if it's a full sentence.
- Captions use .Code in Text and other text formatting as needed.
- If a caption refers to colors shown in a figure, query the author to reword to remove them. Figures will appear in color in the electronic version of the book, but (except in rare cases) the book will be printed in black and white, and all parts of the figure must be identifiable without references to colors.
- For information on credit lines in figure captions, see section 22.

14 Lists

Manning books use numbered lists and bulleted lists. There should always be text that leads directly into the list; a list can't immediately follow a heading (exception: the Summary section). If this lead-in text is a complete sentence, it should end with a colon. If the lead-in is a phrase, not a complete sentence, it doesn't get any punctuation. For examples, see section 4.3.

14.1 Numbered lists

Numbered lists are used to explain a sequence of steps in which the order of the steps matters. Numbered steps should use active voice. The reader is progressing logically through the steps, so introductory words such as *now*, *then*, *next*, and *finally* generally aren't necessary. (Author preference may vary.)

If a list includes code for the reader to enter, the typesetter will take care of indenting it correctly. If a step takes

multiple paragraphs, use the .List Body style for the continuation paragraphs. And if a list item includes substeps, use the .List Numbered Sub style.

Here's an example:

1. Choose File > Save As. **[.List Numbered]**
2. In the Save As dialog box, do the following:
    a. Enter a filename for your document. **[.List Numbered Sub]**
    b. Choose Word Document from the Save as Type list.

Then click OK. **[.List Body]**

TIP If Word gets mixed up and starts a numbered list with any number other than 1, right-click in the new list and choose Restart at 1.

14.2 Bulleted lists

Bulleted lists are used when the sequence of the points isn't critical. If a bullet point has a lead-in, make it *italic*. If the lead-in is a phrase, follow it with an em dash (—) and the text of the bullet beginning with an uppercase letter. If the lead-in phrase is a sentence, follow it with a period and a space. If a bullet's text includes more than one paragraph, use the .List Body style for paragraphs after the first. You can also create sub-lists with the .List Bullet Sub style. Here's an example:

- *Cats are smarter than dogs.* Also keep these facts in mind: **[.List Bullet]**
    - *Cats*—Felines that have babies called kittens **[.List Bullet Sub]**
    - *Dogs*—Canines that have babies called puppies

  United States citizens own more dogs than cats. **[.List Body]**
- *Dogs are smarter than turtles.*

15 Callouts and sidebars

Manning uses two kinds of page element to present information that's related to, but not directly part of, the current topic of discussion: callouts and sidebars.

15.1 Callouts

A *callout* is a brief aside with these characteristics:

- It has **one of these four headings: Note, Tip, Warning, and Definition**. (If an author has used other headings, check with the production editor to see whether to query them.)
- It's **one paragraph or less**.
- It *does not* include lines of code, figures, or lists.

Here is an example:

NOTE The callout's heading uses the **.Callout Head** style and indicates the type of callout. It's separated from the callout's text by a space. The rest of the paragraph uses the **.Callout** style.

15.2 Sidebars

If an aside needs a longer heading, is longer than a paragraph, or includes code snippets or figures, it should be a sidebar rather than a callout. A *sidebar* presents more information than a callout, generally on a topic that's tangentially related to the subject. Sidebars can have longer, descriptive headings (written with the **.Sidebar Head** style).

Sidebars can include code snippets, figures, and lists. They can also include footnotes (see section 18.3). Figures within sidebars aren't numbered, but they do need a caption. Use the same styles for these elements that you use elsewhere. Following the last paragraph of the sidebar, add an empty paragraph and format it with the **.Sidebar End** style.

Here's an example:

Sidebars can have long, descriptive headings

This is an example of a sidebar. It can have more than one paragraph, and it can also include bulleted and numbered lists, as well as code:

```
<form action="#" method="post">
    <div>
         <label for="name">Text Input:</label>
         <input type="text" name="name" id="name" value="" tabindex="1" />
```

If you use a figure in a sidebar, it should have a caption but no number.



This figure is part of the sidebar, so it has a caption but no number.

The last paragraph of a sidebar should be blank and formatted with .Sidebar End.

16 Cross-citing book elements

References to figures, tables, listings, chapters, parts, and appendixes within the book should be lowercase. For example, "See figure 5.9" or "We'll discuss this further in chapter 7."

16.1 Figures and tables

Figures and tables must always have a numbered call out in the text so the reader can easily locate the correct figure/table on the page if it doesn't immediately follow its reference. For example, "Next, add a widget to the layout; see figure 3.1" or "Next, add a widget to the layout (figure 3.1)."

- In the manuscript, place the figure/table and its caption *immediately after the paragraph that refers to it*. During layout, the typesetter often needs to move figures/tables on the page for design or space reasons, but they will keep those elements as close as possible to their location in the manuscript.
- If a one-sentence paragraph introduces a figure/table, combine it with the surrounding text.

16.2 Code listings

Code listings, unlike figures and tables, always follow directly after the paragraph in which they're referenced. For this reason, a numbered reference in the text isn't always needed—for example, it's extraneous to say "as shown in listing 1.5" if the listing comes immediately after those words.

- If a listing reference is in the sentence immediately before the listing, replace it with "as in the following listing" or "as follows" or similar. (*Do not* use the words *above* and *below*.) It's also fine not to specifically mention the listing at all if the preceding text is clearly leading up to or explaining what will be shown in a code example ("Next we need to write the `BuildWidget()` class.").
- If a listing reference is more than one sentence and more than three lines removed from the listing, leave the numbered reference in place.

16.3 Chapters, sections, and other parts of a book

- Chapter titles should not be included ("We'll discuss this in detail in chapter 3.").
- Headings should not be included to numbered (level-1 and level-2) sections ("For more information, refer to section 5.2.3."). Exception: *Do* use the heading when referring to level-3 sections.
- References to a preface, appendix, foreword, index, glossary, or other section of a book are not capitalized.

17 References and citing sources

17.1 Books, journal article, and other printed materials

The decision of where to place reference source information depends primarily on author preference. Source details can appear

- In the text or parentheses
- In footnotes (see section 18)
- As academic citations/reference list:
    - Author/date citations
    - Numbered citations

Do not make major changes to citation/reference formatting or placement without checking with your production editor and/or Alisa (alla@manning.com) first.

IN-TEXT REFERENCES

References in text to external books and articles should provide enough information that readers can find the source. For example:

- To learn more about this, see JJ Geewax's *API Design Patterns* (Manning, 2021).
- To learn more about this topic, check out *C++ Concurrency in Action*, 2nd ed., by Brian Williams (Manning, 2019).

Books noted in the text should be in title case and italicized. Articles should be in title case and surrounded by quotes:

- *Book*—To learn more about this, see JJ Geewax's *API Design Patterns* (Manning, 2021).
- *Article*—To find out more, be sure to check out Erik Voorhees's "It's All About the Blockchain" (https://www.investopedia.com/terms/b/blockchain.asp).

ACADEMIC-STYLED REFERENCES

Academic-styled references can appear either in footnotes (see section 18) or a reference list. If using a reference list, citations can use a numbered or author/date format:

- *Numbered citations*—If the author numbered the reference citations in the text, the reference list at the end of the book should have a section for each chapter with that chapter's references in numerical order starting with 1. In the chapter, put the citation numbers in square brackets inside punctuation: "This technique was first used by Smith" [4]. If more than one citation is listed, put them all in the same brackets. For example: [1–3, 5].
- *Author/date citations*—Citations can use author name(s) and year in parentheses: "This technique was first used by Smith (2021)" or "Next we present the New Widget technique (Jones and Johnson, 2021)." If there are three or more authors, use "et al.": "Recent research supports this theory (Smith et al., 2022)." The reference list is placed at the back of the book in alphabetical order.

> **NOTE** The reference list should be placed at the end of the book. However if the author has a strong preference, a numbered reference list can appear at the end of each chapter. If so, place the reference list after the Summary section.

Most authors who use academic-style references have a preferred format for the source information (Chicago Manual of Style numbered or author/date, Harvard, Vancouver, IEEE, APA, etc.). **Whatever reference style/format the author has used is fine, but make sure it's consistent throughout.** If the author has not used a consistent format, use one of the CMoS formats.

17.2 Websites and URLs

References to websites should provide enough information for the reader to locate the source even if the URL changes: the name of the website, product, or software; an article title; and so on. Follow these guidelines:

- If a URL begins with www, as in www.manning.com, omit http:// from the beginning of the address. If the URL begins with anything other than www, include the http:// or https://, as appropriate (e.g., http://video.google.com)/.
- Long, convoluted URLs in text should be shortened using Manning's URL-shortener tool, for ease of reading and typing. Follow these guidelines to decide whether a URL needs to be shortened:
    - If a URL is in a code listing or snippet, or is being used as part of an example, leave it as is.
    - If the URL begins with http://mng.bz/, it was created at an earlier production stage using the Manning URL-shortening tool. Check to be sure it works, like any other URL.
    - If the URL is longer than about 50 characters and/or contains lots of random letters and numbers, it needs to be shortened.

  You can request shortened URLs from your production editor.

17.3 Quotation sources

- Any quotation from a source that is not in the public domain and that is more than a few sentences long requires permission from the original author/source. If an author uses long quotations from recent works and doesn't credit the source, check with Alisa (alla@manning.com) about whether permission is needed.
- Every quotation should be accompanied by source information that is as complete as possible, including the name of the speaker/author and where the quotation originally appeared. If an author repeatedly cites online quotation websites as sources (BrainyQuotes and similar), query the Production Editor about whether to request more reliable and specific sources.

NOTE It's Manning policy not to use Wikipedia as a source and not to use images from Wikipedia. For more details, see section 23.

18 Footnotes

In some situations, footnotes are allowed in Manning books. The basic rules for using footnotes are as follows.

18.1 Footnotes allowed

Authors may use footnotes for references, including weblinks, books, journal articles, online resources, etc. While Manning does not require that a certain style be followed, all reference footnotes should be consistently formatted and abide by the following rules:

- All footnotes should end with a period.
- If an author uses footnotes for only URLs, leave as is.
- All reference-style footnotes should include as much information as available, such as the author name, book/article/page/entry title, date published/posted, main website name, URL, etc. The footnote doesn't need to include every bit of information, but it should contain enough to be easily found by the reader:

  [1]See John Gruber, "Markdown: Basics," *Daring Fireball*, http://daringfireball.net/projects/markdown/basics.

  If a reference does not include enough information to be easily located, query the author to provide
- If the author cites references in the text, leave as is. Do not move to footnotes. If you are unsure of how to manage footnote use, check with Alisa (alla@manning.com).
- Follow author for capitalization/punctuation but be consistent. For example, any of the following is acceptable:
    - Martha Godlin and Peter Winston. "Gaming's New Face: The Professional Player Is King." *Eurogamer*, 2022. https://www.eurogamer.net/Gamings_New_Face.
    - Martha Godlin and Peter Winston. Gaming's new face: The professional player is king. Eurogamer, 2022. https://www.eurogamer.net/Gamings_New_Face.

- Martha Godlin and Peter Winston, Gaming's new face: the professional player is king, *Eurogamer*, 2022. https://www.eurogamer.net/Gamings_New_Face.

Consistency is key!

- Footnotes can be styled as .Body 1 for readability if needed. The typesetter will have to manually format them so style isn't critical.

18.2 Footnotes not allowed

Footnotes should not be used for informational text. These footnotes should be converted to a callout or sidebar. However, q*uery the author before changing them*. If the author insists on using footnotes for informational text, check with Alisa (alla@manning.com) for guidance on how to proceed.

Incorrect:

[1]We know odd numbers are problematic, but is there an even number without a solution as well? (hint: 10)

Correct:

NOTE We know odd numbers are problematic, but is there an even number without a solution as well? (hint: 10)

18.3 Footnotes in sidebars

Unlike footnotes in the text, footnotes in sidebars are manually lettered (beginning with *a)* rather than numbered and are placed immediately after the sidebar text. If another sidebar in the chapter has footnotes, the next set starts over with the letter *a*.

Cracking

Since a crack involves intricate hacking in assembly (imposed constraints), the intros too started involving creative and often-undocumented tricks with the equipped CPU and GPU (self-imposed constraints). Sometimes, the crack intro was even more complex than the cracked software. Crackers sure found a clever way to show off their hacking skills!

Eventually, cracking evolved into what Eurogamer calls an "interactive art experience," thankfully leaving the illegal part behind.[a] The demoscene subculture comes with many implicit rules, such as emphasizing originality (what they label creativity) over ripping works or assets of others.

[a]Dan Whitehead. Linger in shadows: Scene but not heard. *Eurogamer*, 2008. https://www.eurogamer.net/articles/linge_r-in-shadows-hands-on.

19 Equations

Some books include equations and mathematical expressions within the body of the text (inline) or standing alone between paragraphs (display equations):

- **Never change equations**. If you see what appears to be an error, inconsistency (e.g., italics vs. nonitalics), or typo, always query the author.
- If you have any concerns about how to handle equations, check with Alisa (alla@manning.com) for guidance.

19.1 Standard query

In the first chapter that includes equations, the **copyeditor** should include the following standard query:

> AQ: All equations have been retained per the original manuscript. Please confirm that they appear correctly and have been formatted properly throughout. Display equations will be re-created prior to typesetting and will be available for final review in first pages.

In the first chapter that includes equations, the **proofreader** should include the following standard query:

AQ: Please carefully review all equations and confirm that they are correct as set.

NOTE For simplicity, this section uses the term *equation* to refer to any mathematical or scientific expression made up of letters, numbers, and/or symbols.

19.2 Inline vs. display

How to handle equations in Word depends on how they were created. Equations created in Equation Editor do not transfer correctly to the typesetting program, so it is important to differentiate them. Section 19.3 provides directions for identifying how an equation was created.

INLINE

Simple inline equations are those that appear in a line of text and are set in Word:

- Example: "We will use $f(w_1,w_2)$, $a^2 + b^2 = c^2$ for our calculation."
- If a simple inline equation is set in Equation Editor, re-create it in Word using the appropriate styles and delete the equation editor version. *Doublecheck that the Word version appears exactly like the original (doesn't have to be in the same font)*. Also check to be sure the formatting uses Manning styles, as explained in section 2.2.
- If the chapter/book has many or all simple inline equations set in Equation Editor, let your production editor know immediately.

Complex inline equations appear in a line of text but are too complicated to be set in Word. They appear in the text either in Equation Editor or as a graphic:

- Examples: The equations $\nabla f(\boldsymbol{w}_t)$, and y^s_{pred} cannot be set in Word.
- Complex inline equations are re-created in-house. You don't need to do anything to them.
- If the book you are editing has complex inline equations, let your production editor know.

If you are unsure whether an inline equation is simple or complex, check with Alisa (alla@manning.com) for guidance.

DISPLAY

Simple and complex display equations appear on their own line and are re-created in-house. You don't need to do anything to them:

- Simple display example:

$$F_{t+1}(\boldsymbol{x}) = F_t(\boldsymbol{x}) + \alpha_t \cdot h_t(\boldsymbol{x})$$

- Complex display example:

$$Pred = \frac{\partial \mathrm{f}_{\mathrm{loss}}}{\partial h}(y, h(\boldsymbol{x})) = -(y - h(\boldsymbol{x}))$$

If the book you are editing has display equations, let your production editor know.

NUMBERED DISPLAY EQUATIONS

Numbering of equations is optional—do not add if not there. If the author includes an equation number, style it as .Equation Caption on the line under the equation. (It will be moved to the same line during typesetting.):

$$f_{\mathrm{loss}}(y,\boldsymbol{x}) = \frac{1}{2}(y - h_{\boldsymbol{w}}(\boldsymbol{x}))^2 = \frac{1}{2}(y - w_1 x_1 - w_2 x_2)^2 \tag{2.11}$$

- Numbered display equations should be referred to in the text as follows:
  - Equation 2.11 shows the relationship between *x* and *y*.
  - This method shows the relationship between *x* and *y* (equation 2.11).
  - In equation 2.11, we show the relationship between *x* and *y*.
- Callouts for numbered equations are not required.

19.3 Determining how an equation was created

When you come to an inline or display equation, click on any part of it. If the equation was created using regular text in Word, your blinking cursor will appear wherever you click, as it does when you click in any other text. The following equation was created in Word, and the vertical line is the cursor:

$$a^2 + b^2 = | c^2$$

But if the equation was created using Equation Editor or some other tool, when you click it, a box will appear around the entire equation. This equation was created using Equation Editor:

$$(x+a)^n = \sum_{k=0}^{n} \binom{n}{k} x^k a^{n-k}$$

And this one was created in another tool, saved as a graphic, and pasted into Word:

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L} \right)$$

If the book you are editing contains any complex equations (set in Equation Editor or as a graphic), let your production editor know.

20 Exercises

Authors sometimes include exercises for readers to work on. These are generally presented either as single exercises in sidebars scattered through a chapter or as a list of exercises at the end of a chapter (before the Summary section).

Exercises in sidebars usually appear right after the author discusses whatever the topic is. The sidebar heading is the exercise number:

Exercise 2.1

How would you use the Widgetizer script to create a new widget?

Lists of exercises usually come directly before the Summary section and have the unnumbered heading "Exercises."" **Exercises should appear *before* the Summary section.** They're formatted using numbered (.List Numbered) and lettered (.List Numbered Sub) lists:

1. How would you use the Widgetizer script to create a new widget?
    a. By copying and pasting
    b. By opening a new instance
    c. By switching to a new directory

Answers to exercises generally appear in an appendix. They can also appear in the chapter itself as a section titled "Answers," but they must come before the Summary section.

21 Editing graphics

At some point during the copyediting process, you will be notified by email that the graphics (figures) for the project are ready for editing. If you are coming close to finishing your copyedit and have not received the graphics, let your production editor know.

When editing graphics, be sure that the graphic, the figure caption, and the surrounding text align in terms of spelling, terminology, abbreviations, etc. In some cases, you will need to open the corresponding

chapter file. Check the figure against the text and figure caption. For example, if the text and figure caption refer to a *dataset*, the figure should not use *data set*.

Keep in mind the following:

- Use the Tools > Comment option in Adobe Reader to make changes/add comments.
- If a chapter doesn't require any changes to the graphics, put a note on the first page: "GE: No changes."
- If a chapter requires a query to the author, place in a text box and use red font for the query.
- Be sure capitalization is consistent in a figure and across figures. For example, if figure 1 uses "Title Case" for headings and figure 2 uses "Sentence case" for headings, make consistent.
- Treat notations in figures like annotations: use a period for full sentences; no period for nonsentences.

22 Image credit line

If you come across an image that appears to need a credit line, please query the author to provide. Use your best judgment but lean toward over- rather than underquerying. The query can be as simple as "AQ: Credit line?" Check the handoff document for any information on graphics.

22.1 When a credit line is needed

The following images do not need a credit line (remove if present):

- Public domain
- Author created (most graphics in Manning books are author created)
- Screenshots from software

The following images **do** need a credit line (query author):

- Any image from a third party, including photos, images, and charts, graphs, and tables taken directly from another work
- Screenshots from websites

If an image is credited to Wikipedia, change the credit line to the underlying source (see section 23). **If a project has more than a few graphics credited to a Wikipedia page, contact your production editor.**

22.2 Style for credit lines

Credit lines should appear at the end of the figure caption in parentheses with no period. For images and graphics, use "Source"; for photos, use "Photo courtesy of" or "Photo ©" as appropriate). See the following examples:

> Figure 3.3 Members of the LucasArts crew pose for "Ron Gilbert Day." (Photo courtesy of the International House of Mojo)
>
> Figure 2.1 Niklas Luhmann in his home office, consulting his notes. (Photo © Michael Wiegert-Wegener/University Archives Bielefeld)
>
> Figure 10.20 A surface, a tangent plane, its normal vector, and a pair of principal curvatures (Source: Eric Gaba. Licensed under CC BY-SA)
>
> Figure 1.1 Blockchain is increasing being discussed at IT conferences, roundtables, and forums. (Source: https://en.bitcoin.it. Licensed under CC BY 3.0)

If you have any questions about how to style a credit line, contact Alisa (alla@manning.com) for guidance.

23 Wikipedia

23.1 Citing Wikipedia as a source

Authors *should* **not** *cite Wikipedia as the source* of information. If they cite Wikipedia, query them to provide another source or cite the underlying source as footnoted in Wikipedia:

- **Incorrect** (cites Wikipedia page):

  Blockchain is also being used in peer-to-peer energy trading (https://en.wikipedia.org/wiki/Blockchain).

- **Correct** (cites underlying source as footnoted in Wikipedia):

  Blockchain is also being used in peer-to-peer energy trading (Erik Voorhees, "It's All About the Blockchain," *Money and State,* October 30, 2015).

NOTE If an author insists on citing Wikipedia itself or quotes directly from a Wikipedia article, let your production editor know.

23.2 Using images from Wikipedia

Authors can use images taken from Wikipedia, but *they should* **not** *cite Wikipedia as the source*. Instead, the underlying source should be used:

- **Incorrect** (cites Wikipedia page):

  Figure 1.1 Blockchain is increasing being discussed at IT conferences, roundtables, and forums (https://en.wikipedia.org/wiki/Blockchain)

- **Correct** (cites underlying source):

  Figure 1.1 Blockchain is increasing being discussed at IT conferences, roundtables, and forums. (Source: https://en.bitcoin.it. Licensed under CC BY 3.0)

If an author cites a Wikipedia page as the source for an image, change it to the underlying source. To identify the underlying source, click the image on the Wikipedia page. If not all information is present in the popup window, click More Details. Scroll down to the Summary to find permission and source/author details.

If a project has more than a few images credited to Wikipedia, let your production editor know.

24 Breaking URLs and code elements

This section applies to chapters that have been typeset (proofreading only). Manning books contain many code words and phrases in the body of the text, as well as filenames and URLs. To avoid confusing readers with extra hyphens, follow these rules for hyphenating or breaking these elements on the printed page:

- Break code elements, URLs, and filenames in the text only if absolutely necessary.
- Break *before* a period, question mark, or exclamation point.
- Break *after* a slash, number sign, ampersand, or underscore.
- Do *not* break after a hyphen, because it will cause confusion for readers—they won't know if it's an actual hyphen in the URL or one added by default at the line break.
- If a code element in the text consists of multiple linked words, each of which is capitalized (such as `MethodInvoker`), hyphenate between words—not in the middle of a word.
- If the code element is a single word (such as `IEnumerable`), don't break it.
- Don't break in the middle of a word unless absolutely necessary. If there is no way to break around a symbol, then break between letters, but do *not* add a hyphen.

Here are some examples of how to break code elements:

- `IEnumerable`—Don't break.
- `Function`—Don't break.
- `MethodInvoker`—Hyphenate as follows: `Method-Invoker`.
- `System.Predicate<T>`—Break before the period: `System .Predicate<T>`.

And here are some examples of how to break—and not break—URLs.

Yes:

- www.amazon
  .com/
- www.amazon.com/Chicago-Manual-Style-University-Press/dp/
  0226104036
- www.amazon.com/Chicago-Manual-Style-Univ
  ersity-Press/dp/0226104036

No:

- http://www.amazon.
  com/
- http://www.amazon.com/Chicago-Manual-Style-University-
  Press/dp/0226104036/
- http://www.amazon.com/Chicago-Manual-Style-Uni-
  versity-Press/dp/0226104036

25 Project-specific and standard word lists

- A project-specific word list (style sheet) should accompany the manuscript from the copyeditor to the proofreader. Consistency is the primary rule to be followed in treatment of words as well as in determining which words and expressions should be set in code font.
- Author preference may vary. Be sure to note the usage in the book's word list, and be sure it's consistent throughout the book.
- Any time you're not sure about closing up a prefix or whether a term is one word or two, if it's not in the following list, check the *Chicago Manual of Style or* Merriam-Webster online.
- To check the capitalization and format of product names, languages, and so on, always check the technology's home page rather than relying on blogs or number of Google hits.

- 24/7
- 2D (3D) or two-dimensional (three-dimensional) but *not* 2-dimensational (3-dimensional)
- a.m. or A.M.
- above—Do not use to point to previous text; change to "previous" or the like
- acknowledgments
- ad hoc (adj, adv)—No italics or hyphen
- add-on (n, adj),
  add on (v)
- aka
- Alt key
- Alt-N
- anti (prefix: closed unless the root word starts with *i* or a capital letter)
- app—OK on first use for iPhone apps and similar; OK on second use as abbreviation for *application*
- appendixes
- `async` (code font)
- autogenerate
- backend (n, adj)
- back up (v), backup (n)
- backquote
- backslash
- Backspace key

- backtick
- bash shell
- below—Do not use to point to later text; use "following" or the like
- BIOS
- bitwise operators
- Boolean (unless referring to a datatype in code, in which case `boolean`)
- boot loader
- braces or curly braces
- brackets or square brackets
- breadcrumb navigation
- browsable (n)
- built-in (adj, n)
- Caps Lock key
- check box
- checkout (n, adj), check out (v)
- class path *but* `CLASSPATH` environment variable
- click (not click on)—Verb to use with onscreen options
- clickthrough (n, adj), click through (v)
- client side (n), client-side (a)
- client/server
- command line (n), command-line (a)
- compile time (n), compile-time (a)
- criterion (s), criteria (p)
- cross reference (n), cross-reference (v),
- cross section (n),
  cross-section (v, adj)
- crowdsource
- Ctrl key
- Ctrl-Alt-Delete
- Ctrl-N
- curly braces or braces
- data is
- data set or dataset (be consistent)
- data type or datatype (be consistent)
- decision-making
- deep learning (compound noun; no hyphen when used as an adj)
- design time (n),
  design-time (a)
- double-click (n, v)
- double-precision (a)
- down arrow
- drop-down (a)
- e-book
- e-business
- e-commerce
- email
- Enter key
- equals sign
- Esc key
- failback (n)
- failover (n)
- file server
- file type
- filename
- filesystem
- follow up (v),
  follow-up (adj, n)
- frontend (n, adj)
- ft (abbrev. for *feet*)
- functional programming (compound noun; no hyphen when used as an adj)
- GB (gigabyte),
  Gb (gigabit),
  GBps (gigabytes/sec),
  Gbps (gigabits/sec)
- GHz
- grayscale
- hardcoded
- high-level (adj)
- hit—Do not use as a verb with keyboard keys; use *press*
- hostname
- hotspot
- in. (abbrev. for *inch*)
- inline
- inode
- internet
- Internet Explorer
- Internet of Things (IoT)
- internetwork

- intranet
- JAR—Java Archive
- JavaScript
- KB (kilobyte),
  Kb (kilobit),
  KBps (kilobytes/sec),
  Kbps (kilobits/sec)
- keycode
- keymaps
- keypad
- keystroke
- keywords
- kHz (kilohertz)
- leftmost
- leveled (not levelled)
- life cycle
- list box
- log in, log out, or log on (v); login, logout, or logon (n or adj)
- lookup (n), look up (v)
- lowercase
- lower-level (a)
- machine learning (compound noun; no hyphen when used as an adj)
- macOS
- markup (n), mark up (v)
- mashup (n, adj),
  mash up (v)
- MB (megabyte),
  Mb (megabit),
  MBps (megabytes/sec),
  Mbps (megabits/sec)
- menu bar
- metacharacter
- metadata
- MHz (megahertz)
- microservice (n), microservice-based (adj) *but* microservices architecture
- mockup
- multi (prefix: closed except before vowels or unless it makes the word unclear)
- multiline
- multitasking
- multithreaded
- multitier
- multiuser
- namespace
- newline character
- non (prefix: closed except before vowels, or unless it makes the word unclear, with a few tech exceptions)
- non-blocking
- nonlocal
- non-modular
- non-null
- nonstandard
- nonzero
- offline
- offload
- offscreen
- onscreen
- open source (n or a)
- p.m. or P.M.
- pagefile
- pandas (always lowercase)
- parentheses (p), parenthesis (s)
- passcode
- passphrase
- pathname
- percent—Use the % sign.
- PhD
- plug-in (a, n) or plugin (be consistent)
- pop up (v),
  pop-up (n, a)
- post (prefix: closed unless the root word starts with a capital letter)
- pre (prefix: closed unless the root word starts with an *e* or a capital letter)
- press—Verb to use with keyboard keys
- progress bar
- pseudoattribute
- pseudocode
- pseudoterminal
- pull-down (a)
- random-access (a)

- re (prefix: closed except before vowels or unless it makes the word unclear)
- read/write
- read-only (a)
- real time (n),
  real-time (a)
- redirection
- Return (key)
- right-click
- roll back (v),
  rollback (n)
- run time (n), run-time (adj): program lifecycle phase. *But* runtime environment, runtime library, runtime system, etc. to refer to the environment in which a program runs
- screen dump
- screenful
- screensaver
- screenshot
- scrollbar
- server side (n),
  server-side (a)
- shell (lowercase even in a shell name: *Bourne shell*)
- Shift key
- spacebar
- spell-checker
- split screen
- square brackets or brackets
- standalone
- startup file
- status bar
- stylesheet
- subprocesses
- Tab key
- Telnet (the protocol), *but* telnet (v)
- text box
- text-input mode
- the web (n); *but* Web 1.0, Web 2.0
- there is / there are: reword when possible
- third party (n),
  third-party (a); *not* 3rd
- this is: reword when possible
- thread pool (n),
  thread pooling (n)
- thread safe
- time zone
- time-sharing processes
- timestamp
- toolbar
- toolkit
- tooltip
- top-level (a)
- toward
- tradeoff (no hyphen per Marjan)
- troubleshoot
- United States (n),
  US (adj)
- UNIX
- up arrow
- uppercase and lowercase
- upper-left corner
- user ID
- username
- versus in text; vs. in headings, table captions, figure labels, and code annotations
- vice versa
- web (a, n)
- Web 1.0, Web 2.0
- web client
- web page
- web server
- web service *but also Web Services*
- webcast
- webmaster
- website
- white pages
- whitespace
- who and whom: follow the strict grammatical rules (e.g. "Whom is this book for?" rather than "Who is this book for?")
- wiki
- wildcard
- wizard (a, n), *but* Wizard (in the name of a specific wizard)

- workaround
- workgroup
- workstation
- World Wide Web
- wraparound
- writable
- write-only (a)
- zeros
- ZIP code
- zip file

26 Abbreviations

The use of abbreviations can vary from book to book based primarily on the audience and author's preference. This section should be approached as a guideline, not as a set of rules etched in stone. Use your best judgment. The following are general rules for using abbreviations:

- Add all abbreviations to the book's style sheet.
- Define once per chapter, unless it creates unnecessary repetitiveness (e.g., a book on Amazon Web Services does need to define AWS every chapter).
- It is OK if an author defines an acronym and then never uses the term or the acronym again. The definition may be introducing the reader to new terminology.
- If an acronym stands for a noun, make the plural by adding an *s* (no apostrophe). Example: GPUs, not GPU's.
- If an author defines an acronym and subsequently uses the spelled-out term, replace the spelled-out terms with the acronym. Example: the author defines "virtual machine (VM)" but continues to use the full term in the text; replace "virtual machine" with "VM" where it occurs.
    - Exception: Abbreviations can be redefined in the Summary section.
    - Exception: The author is intentionally redefining the term.
- Some authors use many acronyms; others use few or none. Don't add acronyms the author hasn't used, even if a term has been written in acronym form in other books. Example: the author discusses virtual machines but never uses the common VM acronym; leave it as is or query the author if you wish.

26.1 To spell or not spell (out)

Determining whether to spell out a term is based on the following:

- Abbreviations should be spelled out on first use, although some universally known abbreviations never require definition. Section 26.3 provides some common abbreviations that do not need to be spelled out. However, if an author spells them out, leave as is.
- Section 26.4 provides some common abbreviations that usually need to be spelled out. However, for certain audiences, they may not need to be defined. Query the author if you have any doubts.

26.2 Abbreviations in headers/titles and This chapter covers

The rules for using abbreviations in headers/titles and the **This chapter covers** section are as follows:

- Do not define abbreviations in headers or chapter titles:

    Incorrect: 1.2 Using content management systems (CMS)
    Correct: 1.2 Using content management systems

    Instead, define the abbreviation on the first use in the main text.
- If an abbreviation appears in a header, be sure it has been previously defined. If it hasn't, spell it out.
- Do not allow headers made up only of an abbreviation:

    Incorrect: 1.2 FTP
    Correct: 1.2 File transfer protocol
- It's OK to use a nonstandard abbreviation in the chapter title and **This chapter covers** that has not yet been defined, as long as it's defined in the text.

26.3 Standard abbreviations

The following is a nonexhaustive list of abbreviations that do not need to be spelled out. If you are unsure whether an abbreviation needs to be spelled out, query the author (also see section 26.1) or check with Alisa (alla@manning.com).

2D	GB*	JPQL	SDK
3D	GHz*	JSON	SOLID
AI	GNU	KB*	SQL
API	gRPC	KHz*	SSD
ASCII	GUI	LAN	SSH
BIOS	HTML	MB*	SSL
CAD	HTTP	MHz*	TB*
COBOL	HTTPS	NoSQL	UI
CPU	IDE	OS	URL
CSS	I/O	PC	XML
CSV	IP	PHP	YAML
DevOps	ISDN	PL/SQL	
DLL	ISP	REST	
FTP	IT	RESTful	

*Spell out if not used with a number.

For the following terms, use the abbreviation and spell out in parentheses:

ACID	atomicity, consistency, isolation, durability
ETL	extract, transform, load
CRUD	create, read, update, and delete
REPL	read–evaluate–print loop
WYSIWYG	what you see is what you get

26.4 Nonstandard abbreviations

The following are two (nonexhaustive) lists of abbreviations that should be spelled out on first use (see section 26.1). Spelling out may not always be necessary, depending on the target audience and author preference. Use your best judgment and be consistent.

AMAZON ABBREVIATIONS

Amazon EBS	Amazon Elastic Block Storage
Amazon EFS	Amazon Elastic File System
Amazon EC2	Amazon Elastic Cloud Computing
Amazon ECS	Amazon EC2 Container Service
Amazon ELB	Amazon Elastic Load Balancer
Amazon EMR	Amazon Elastic Map Reduce
Amazon RDS	Amazon Relational Database Service
Amazon S3	Amazon Simple Storage Solution
Amazon SNS	Amazon Simple Notification Service
Amazon SQS	Amazon Simple Queue Service
Amazon SWF	Amazon Simple Workflow Service
AMI	Amazon Machine Image
ARN	Amazon Resource Name
AWS	Amazon Web Services

NOTE *Amazon* should be part of the abbreviations. However, use your best judgment. For example, if an entire chapter discusses Amazon S3 and Amazon ECS, it's not necessary to use *Amazon* in each instance. However, change *AWS* to *Amazon* (e.g., AWS EC2 should be Amazon EC2)

OTHER NONSTANDARD ABBREVIATIONS

1NF	first normal form
2NF	second normal form
3NF	third normal form
ACL	access control list
AD	active directory
AES	advanced encryption system
AJAX	asynchronous JavaScript And XML
ALB	Application Load Balancer
AR	augmented reality
ASP	Active Server Pages
AT	acceptance testing
AZ	availability zone
BI	business intelligent
BDD	behavior-driven development
CBSP	cloud-based security providers
CDN	content delivery network
CI/CD	continuous integration/continuous delivery
CIO	chief information officer
CMS	content management system
CLI	command-line interface
COM	component object model
CORS	cross-origin resource sharing
CQRS	command query responsibility segregation
CQS	command query separation
CSP	content security policy
CSRF	cross-site request forgery
CVS	concurrent versioning system
DAO	data access object
DBA	database administrator
DBMS	database management system
DDD	demand-driven design
DDoS	distributed denial of services
DES	data encryption standard
Dev	development
DML	Data Manipulation Language
DMS	database management system; database migration service
DNA	digital network architecture
DOM	document object model
DOS	denial of services
DQL	Data Query Language
DTO	data transfer object
EE	Enterprise Edition
ERM	entity relationship model
ERD	entity relationship diagram
ES	ECMAScript
ETL*	extract, transform, load
FaaS	Function as a Service
FCL	Framework Class Library
FDD	feature-driven development
FIFO	first in, first out
GCP	Google Cloud Platform
GCS	Google Cloud Storage
HPC	high-performance computing
IAM	identity and access management
IaaS	infrastructure as a service
ID	ID
IDN	integrated digital network
IIFE	immediately invoked function expression
IMS	intelligent maintenance systems
IPSec	internet protocol security
ISA	instruction set architecture
ISS	Internet Information Services
J2EE	Java 2 Enterprise Edition
J2ME	Java 2 Micro Edition
J2SE	Java 2 Standard Edition
JDBC	Java Database Connectivity
JDK	Java Development Kit
JFC	Java Foundation Classes
JPA	Java Persistence API
JRE	Java Runtime Environment
JS	JavaScript
JSP	JavaServer Pages
JVM	Java virtual machine
LIBSVM	LIBSVM library
LIFO	last in, first out
MFA	multifactor authentication
ML	machine learning
MLOps	machine learning operations
MVC	model, view, and controller
NAT	network address translation
NLP	natural language processing
NFS	network file system
OCR	optical character recognition
OLAP	Online Analytical Processing
OOP	object-oriented programming
ORM	object-relational mapping
PaaS	platform as a service
PVT	product verification testing
RDS	relational database service; but Amazon Relational Database Service
RDSMS	relational data stream management system
REGEX	regular expression
RPC	remote procedure call
RRS	Reduced Redundancy Storage
SaaS	software as a service
SAP	Systems Applications & Products
SCM	source code management
SOAP	Simple Object Access Protocol
SQLi	SQL injection
SQLXML	SQL Server XML
SSE	server side encryption

SVN	subversion
TDD	test-driven development
TLS	transport layer security
UAT	user acceptance testing
UUID	universally unique identifier
VB	Visual Basic
VB6	Visual Basic 6.0
VBA	Visual Basic for Applications
VM	virtual machine
VPC	virtual private cloud
VPG	virtual private gateway
VPS	virtual private server
VR	virtual reality
VS	Visual Studio
VSS	visual source safe
XP	extreme programming
XSS	cross-site scripting